Aditya Murarka

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SUMMARY

Software Engineer with 5+ years of experience developing React and TypeScript applications, Python and Node.js APIs, and real-time systems. Skilled in micro frontend architecture, WebSockets, event-driven integrations, AI agents, SQL, performance optimization, and CI/CD. Delivered measurable gains in application performance, test coverage, code reuse, and cloud efficiency.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL, Swift, SPARQL

Frontend: React.js, Next.js, Redux, Zustand, React Router, SWR, Tailwind CSS, Material UI, Storybook, HTML5, CSS3, Responsive Design, SwiftUI

Backend & Data: Node.js, Express, FastAPI, Spring Boot, REST APIs, GraphQL, PostgreSQL, Supabase, SQLite, Firebase, Prisma ORM, Flask, OAuth, SSO, JWT, pgvector

Real-Time & AI: WebSockets, Server-Sent Events (SSE), Kafka, SharedWorker, Socket.IO, asyncio, LangChain, Agentic AI, Retrieval-Augmented Generation (RAG), MCP, Groq, scikit-learn, NumPy, Pandas

Architecture & Performance: Micro Frontends, Module Federation, React Server Components, Webpack, Code Splitting, Lazy Loading, Core Web Vitals, Lighthouse, WCAG

Testing & Cloud: Jest, React Testing Library, pytest, Git, CI/CD, GitHub Actions, AWS, Azure, Vercel, Render, ESLint, Ruff, Adobe Analytics

PROFESSIONAL EXPERIENCE

Software Engineer III, Walmart Global Tech **Mar 2022 – Jul 2024**

- Designed and developed the conversational UI for an enterprise RAG assistant, supporting multi-turn natural-language queries over supply chain and inventory data with streaming responses and dynamic rendering of text, tables, and interactive charts.
- Modernized legacy AngularJS screens and JavaScript modules into React and TypeScript within a Module Federation micro-frontend architecture, improving screen performance by up to 30%.
- Engineered a real-time notification system using React, Server-Sent Events (SSE), Kafka, and Spring Boot; leveraged SharedWorker to coordinate connections across browser tabs and synchronize notifications without redundant polling.
- Led an Adobe Analytics proof of concept and 20-screen MVP, guiding instrumentation across 80+ screens; developed supply chain dashboards exposing idle resources and supporting a 15% reduction in cloud spend.
- Performed root-cause analysis and rollback of a failed shared component library release, improving versioning and release coordination practices across teams.
- Raised unit and component test coverage to 93% using Jest, React Testing Library, and API mocking; resolved accessibility findings identified by external WCAG audits.

Software Engineer I, Nineleaps Technology Solutions **Aug 2019 – Mar 2022**

- Led and mentored a five-engineer team delivering SPRY, a sports compliance platform; owned sprint planning, technical scoping, client communication, and releases.
- Developed React and Redux workflows for contracts, multi-step forms, file uploads, and earnings dashboards, integrating frontend features with backend services.
- Built a shared Storybook component library, introduced ESLint and Prettier standards, and organized code into a Yarn Workspaces monorepo, reducing duplicate UI code by 30%.
- Architected a TypeScript-based content management platform using Redux-Saga, React Router, recursive tree navigation, and reusable widgets for non-technical content authors.
- Improved page-load performance through route-level code splitting and bundle optimization; used Lighthouse audits to investigate regressions and guide frontend improvements.

PROJECTS

WeVibe – iOS Dating App – Arizona State University **Jan 2026 – May 2026**

- Built end-to-end product flows across a SwiftUI iOS client and Node.js, Express, and TypeScript backend, including Firebase authentication, onboarding, profiles, photo uploads, location sync, matchmaking, and chat.
- Implemented real-time chat via Socket.IO with centralized state, unread badges, typing indicators, queue recovery, background refresh, and Firebase Cloud Messaging (FCM) notifications.
- Extended REST APIs and Prisma/PostgreSQL services for Apple Sign-In, account deletion, and Gemini-powered bio generation with cooldowns and daily rate limits.

F1 AI – Agentic Race Engineer Assistant – f1-ai-delta.vercel.app **May 2025 – Jul 2025**

- Built an AI-powered Formula 1 race engineer using LangChain and Groq, orchestrating 15 specialized tools for natural-language race analysis, live telemetry, strategy, and ML predictions; implemented dynamic tool selection and error recovery for reliable LLM tool execution.
- Designed a retrieval architecture combining FIA regulation embeddings, Supabase pgvector, and cross-encoder reranking with constrained text-to-SQL, grounding AI responses in cited regulations and historical race records spanning 75+ years.
- Developed a full-stack AI platform using Next.js, FastAPI, and WebSockets, streaming agent execution progress to users and exposing 11 analysis tools to external AI clients through the Model Context Protocol (MCP).

Gotham Watch – Crime Analysis Platform – Arizona State University **Oct 2024 – Dec 2024**

- Built an interactive crime analysis platform with React and dynamic data visualizations, powered by Python and Flask APIs running SPARQL queries over a GraphDB knowledge graph, deployed on Microsoft Azure.

EDUCATION

Master of Science in Software Engineering – Tempe, AZ, US **Aug 2024 – May 2026**
Arizona State University, GPA: 3.63/4.0

- Relevant Coursework: Data Structures & Algorithms, Data Visualization, Cybersecurity, Cloud Computing